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Species range shifts often speed ahead of their modeled climatic niches

Brunno F. Oliveira[a,b,1], Romain Bertrand[c], Malin L. Pinsky[d], Nicolas Casajus[a], Barrett W. Wolfe[e,f], Brett R. Scheffers[g], Fabricio Villalobos[h], Gaël Grenouillet[c], Gretta T. Pecl[e,f], I-Ching Chen[i], J. Alex Baecher[g], Jake A. Lawlor[j], Jennifer Sunday[j], Jérôme Murienne[c], Jonathan Rolland[c], Laura M. Thompson[k,l], Lesley T. Lancaster[m], Madeleine A. Rubenstein[k], Nikki A. Moore[j], R. M. W. J. Bandara[n], Sarah E. Diamond[o], Sarah R. Weiskopf[k], Jonathan Lenoir[p,2], and Lise Comte[q,2]

Affiliations are included on p. 11.

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Anticipating how species distributions will shift with climate change is key for biodiversity conservation and management. Commonly, species' range shifts are observed by analyzing changes in occurrence or abundance data through time, or predicted across different climate change scenarios by modeling species' climatic niches. However, it remains unclear how well these climate-based forecasts align with empirically documented range shifts from monitoring efforts. Here, we tested the congruence between modeled range shifts, predicted using climatic niche models, and documented range shifts, derived from empirical observations collected over recent decades, for more than 9,500 range shifts across over 3,500 marine and terrestrial species. We found that documented and modeled range shifts tend to align in latitudinal direction, with greater alignment for marine (76%) than terrestrial (56%) cases. However, even when the directions aligned, documented shifts exceeded modeled shifts in 62% of cases, nearly twice as often as they lagged behind (38%), and their median rates were four times faster than those of the modeled shifts. Our findings suggest that climate-based models can approximate observed range dynamics under specific conditions, particularly over long time periods and restricted spatial areas, when habitats remain well connected and under low climate fluctuations over time. These insights provide valuable guidance for both improving predictions and informing responses to climate-driven biodiversity redistribution.

niche tracking | distribution shift | model validation | habitat suitability | climate change

Significance

Predicting how and where species will move as climate changes is crucial for protecting biodiversity. Scientists often use niche models based on where species lived historically to forecast shifts in their distributions, but it remains unclear how well these forecasts align with observed changes. By comparing predictions from these models to actual observations of over 9,500 species range shifts across land and sea, we found that while niche models tend to get the direction right, especially for marine species, they often fail to predict the pace of redistribution. Our study highlights when and where niche models work best, where they fall short, and under which conditions predictions capture observations, offering a clearer path for improving forecasts of biodiversity redistribution.

As climate warming accelerates globally (1, 2), species redistribution has emerged as a major challenge for natural resource management and human societies (3–6). Accurate projections of future species distributions under climate change scenarios are essential for effective conservation planning and policy decision-making (7). However, we lack a clear understanding of whether we can reliably predict species redistributions with the accuracy needed to guide these efforts.

Species range shifts have now been documented across diverse taxonomic groups and geographical regions, as populations establish in new areas where climatic conditions become suitable and are extirpated from increasingly unsuitable locations throughout their range (8–10). These range shifts are shaped by a combination of extrinsic factors (e.g., climate exposure and landscape features) (11, 12) and intrinsic species' characteristics (e.g., physiological tolerance and dispersal abilities) (13), which often interact in complex ways (14). As a result, observed species range shifts are far more variable, in both magnitude and direction, than expected from isotherm shifts alone (10, 15), making predictions of climate-driven biodiversity redistribution a major challenge.

Niche models, or species distribution models (SDMs), usually estimate the realized niche as originally defined by G.E. Hutchinson (16)—the occupied portion of the multidimensional environmental (usually climatic) space suitable for the species. These models are widely used to forecast species range shifts under climate change scenarios (17–20), providing valuable insights for natural resource management and conservation planning, but their ability to accurately predict observed range shifts remains uncertain (21–26). Discrepancies between niche-based predictions and observed range shifts can be attributed to four main factors. First, species-climate disequilibrium may arise from biotic interactions (27), historical legacies (28, 29), or microclimatic variation not captured by broad-scale models (30). Second, dispersal limitations, including both external physical barriers and habitat fragmentation, as well as intrinsic constraints on species' dispersal capacity and establishment ability can contribute to the mismatch between niche-based predictions and observed range shifts (31). Third, incomplete representation of key environmental

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[1]To whom correspondence may be addressed. Email: brunno.oliveira@me.com.

[2]J.L. and L.C. contributed equally to this work.

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determinants of species distributions may contribute to the species-climate disequilibrium, as some relevant variables may be missing or poorly characterized. Last, adaptive mechanisms, including both phenotypic plasticity and evolutionary potential (32), can alter the realized niche as estimated by climatic niche models. Understanding these discrepancies can shed light on the mechanisms underlying species' capacity to track shifting climates.

Mismatches between modeled and documented range shifts can also arise from imprecise detection of empirical range shift's observations, as well as underlying model uncertainties. The reliability of empirical estimates is inherently tied to the uncertainty in the observational data (33) and the extent to which underlying models capture relevant ecological processes. These inaccuracies can result from oversimplified assumptions, errors in parameter estimation, or incomplete representations of environmental conditions and ecological factors influencing species distributions (34). Additionally, methodological factors have been identified as primary drivers of variability in documented range shift rates, often surpassing the influence of ecological traits and climate variables (35, 36). For instance, limited sampling frequency, restricted spatial coverage, and short study durations can amplify uncertainty in range shift estimates, potentially obscuring long-term trends (8). These limitations not only affect empirical estimates but also hinder robust evaluation of the accuracy of predictive models in capturing biodiversity redistribution.

Evidence regarding the alignment between modeled and observed range shifts, both in direction and magnitude, is mixed. Many studies report poor agreement in either aspect (24–26, 37, 38), suggesting that niche-based models often fail to accurately predict range shifts under climate change (but see ref. 21). Such inconsistencies highlight the challenges in drawing generalizable conclusions regarding how well species track their climatic niches and thus the ability of niche models to predict range shifts, while also underscoring the need for standardized methodologies to enhance the comparability and reliability of predictions. Previous studies focused either on a restricted spatial extent, usually within a single country (15, 21, 22, 24, 25, 39) or continent (23, 26), or a specific taxonomic group, like birds (23–26), butterflies (22, 25), vascular plants (25), or freshwater fishes (21). There remains a critical need to understand the drivers of these discrepancies across taxa and ecosystems. Addressing this gap would help clarify how extrinsic drivers (e.g., environmental changes, habitat accessibility) (8), intrinsic traits (e.g., dispersal ability, physiological tolerance) (14), and methodological factors (e.g., study duration, latitudinal extent) (35), influence deviations from modeled expectations (i.e., climatic niche tracking), a research topic that remains poorly explored. Ultimately, such efforts would refine our ability to anticipate and manage biodiversity redistribution under climate change.

Here, we evaluate the variation in, and underlying drivers of, the alignment in direction and mismatch in magnitude between modeled and documented latitudinal range shifts across diverse taxonomic groups and geographical areas (*SI Appendix*, Fig. S1). We focus on latitudinal range shifts, as they represent the most pronounced and well-documented geographic response to climate change. We leverage two databases (i.e., BioShifts and CoRE) of range shift observations from the scientific literature covering plants, invertebrates and vertebrates in both the terrestrial and marine realms (40, 41). We develop finely resolved climatic niche models for 3,552 marine and terrestrial species for which 9,539 documented latitudinal range shift estimates were determined (Fig. 1). Each model was built using an ensemble approach and tailored to the metadata of each original study, ensuring that modeled range shifts matched the same geographical extent and temporal window as the documented range shifts.

Results

Overall, our climatic niche models effectively captured the observed distributions for the 3,034 terrestrial and 518 marine species included in our analysis and distributed across several biomes (*SI Appendix*, Fig. S1). The models demonstrated a good balance between sensitivity and specificity, as indicated by an average true skill statistic (TSS) of 0.57 ± 0.13, and a high consistency between predicted suitabilities and observed presences, as indicated by an average Boyce index of 0.95 ± 0.13. While TSS values were statistically similar between realms [terrestrial: 0.57 ± 0.13 SD, marine: 0.57 ± 0.16; ANOVA: $F(1, 3556) = 0.01$, $P = 0.92$], the Boyce index was significantly higher for terrestrial species [terrestrial: 0.98 ± 0.05, marine: 0.81 ± 0.27; ANOVA: $F(11, 3556) = 1{,}015$, $P < 0.001$]. Model performance varied also across taxonomic groups (*SI Appendix*, Fig. S2), with the highest Boyce index values observed for terrestrial plants (Magnoliopsida, Polypodiopsida, Liliopsida), birds (Aves), and insects (Insecta), as well as in marine taxa, for polychaete worms (Polychaeta), crustaceans (Malacostraca), sharks and rays (Elasmobranchii), as well as bony fishes (Actinopteri).

Modeled Range Shifts. The climate suitability of species, estimated by predictions from climate-only niche models tailored to each study period and location as recorded in the BioShifts and CoRE databases (Terrestrial = 8,603; Marine = 936), generally shifted poleward (Fig. 2 and *SI Appendix*, Fig. S3), except for terrestrial species from the Southern Hemisphere which, on average, shifted toward the equator (*SI Appendix*, Fig. S3). These modeled range shifts, estimated from two-dimensional (longitude–latitude) projections, were predominantly oriented along the latitudinal axis, with 67% (±31%) and 62% (±32%) of the predicted shifts in terrestrial and marine systems, respectively, occurring in the latitudinal direction. To enable direct comparison with the documented latitudinal range shifts, we extracted the latitudinal component of each modeled range shift and used these shift values in subsequent analyses.

Alignment and Match Between Documented and Modeled Latitudinal Range Shifts. The direction of modeled latitudinal range shifts (i.e., poleward *vs.* equatorward) aligned with the documented direction in 55.9% and 75.5% of the terrestrial and marine shifts, respectively (*Top-Left* Insets, Fig. 2). This alignment exceeded chance expectations in both realms, based on binomial generalized linear mixed-effects models (GLMMs) with shift direction (alignment *vs.* misalignment) as a binary response variable and an intercept-only structure. Significantly positive intercepts indicated terrestrial shifts were 1.2 times [Intercept = 0.21, 95% CI (0.08, 0.34)], while marine were 3.5 times [Intercept = 1.25, 95% CI (0.98, 1.51)], more likely to align than expected by chance.

Despite the frequent alignment in direction, we found little evidence that the rates of modeled shifts positively relate to the rates of documented shifts in either realm (Fig. 2 and *SI Appendix*, Table S1). These relationships significantly departed from the 1:1 expectation (*SI Appendix*, Table S1), suggesting that niche models failed to reflect the pace of documented range shifts. For directionally aligned shifts, the central tendency of documented range shift rates was nearly four times faster than modeled rates for both marine and terrestrial species, suggesting that climatic niche models tend to underestimate the pace of empirical observations (*SI Appendix*,

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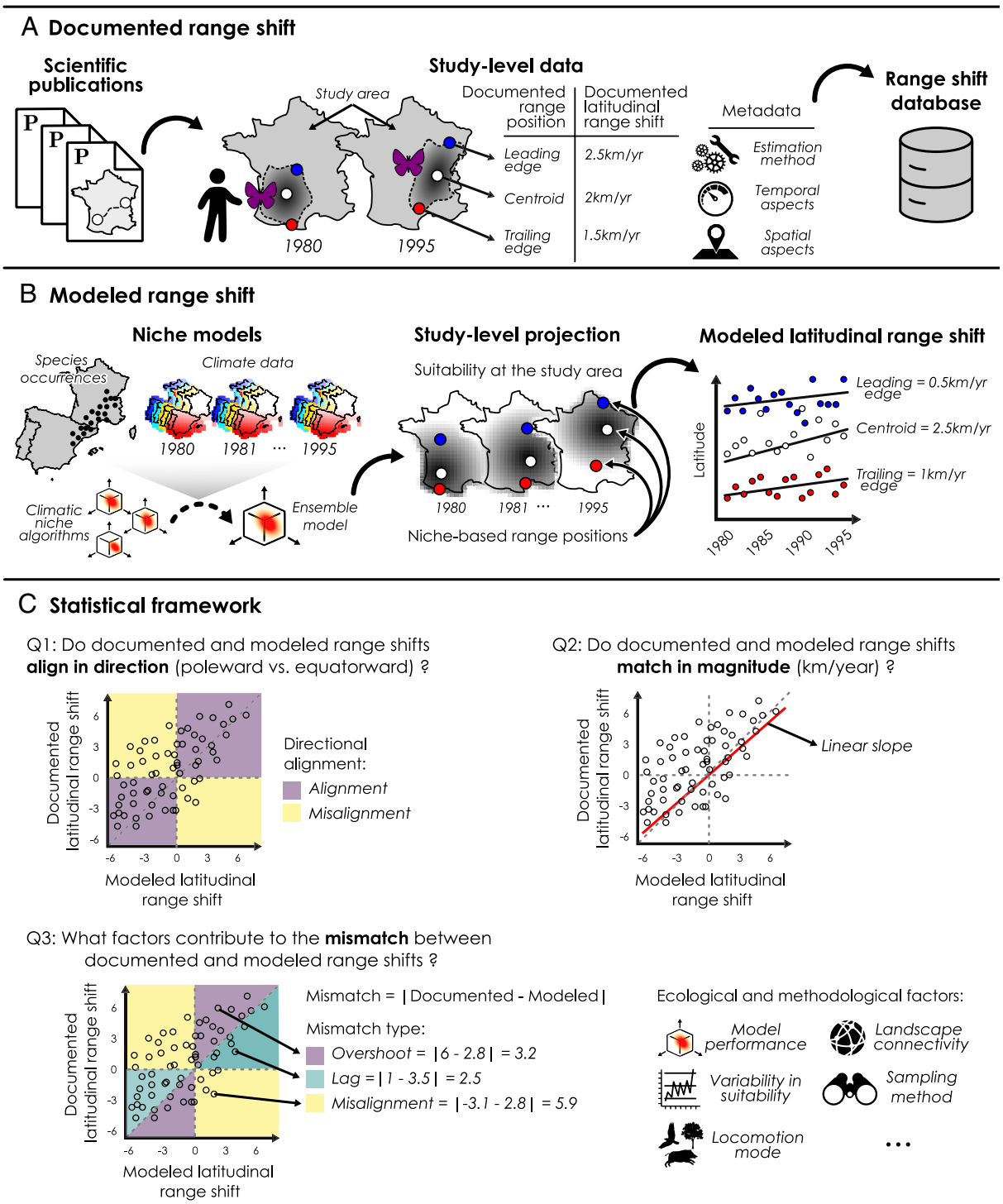

Fig. 1. Graphical overview of the methodological approach used to evaluate the ability of climatic niche models to reproduce documented latitudinal species range shifts. This approach was applied globally to 3,552 species across marine and terrestrial systems. Documented latitudinal range shifts (km/y) at different range positions (*A*) and associated methodological variables were sourced from the BioShifts and CoRE databases, here illustrated with a hypothetical study using a butterfly species in France. To ensure robust comparisons between modeled and documented shifts, we implemented an ensemble modeling framework specifically designed to match the geographical and temporal parameters of each original empirical study. Modeled range shifts (*B*) were estimated through linear regressions to calculate latitudinal changes (km/y) for the centroid, leading edge, and trailing edge of each species range, using yearly suitability maps predicted by niche models. The statistical framework (*C*) included testing the predictive accuracy of modeled range shifts in terms of alignment in direction (Q1) and match in magnitude (Q2) of documented range shifts, and identifying factors influencing the mismatch (Q3) (i.e., absolute differences between documented and modeled shifts), which was categorized as cases of overshoot, lag, or misalignment (see *Methods* for further details).

Table S2). For terrestrial species, modeled predictions underestimated documented shifts (i.e., documented shifts overshoot) in 35.0% (mean 2.23 km/y), overestimated (i.e., documented shifts lag behind) in 20.9% (mean 1.24 km/y), and showed directional misalignment in 44.1% of cases (*SI Appendix*, Table S3). For marine species, modeled predictions underestimated documented shifts in 44.3% (mean 3.41 km/y), overestimated in 31.2% (mean 1.75 km/y), and misaligned in 24.5% of cases (*SI Appendix*, Table S3).

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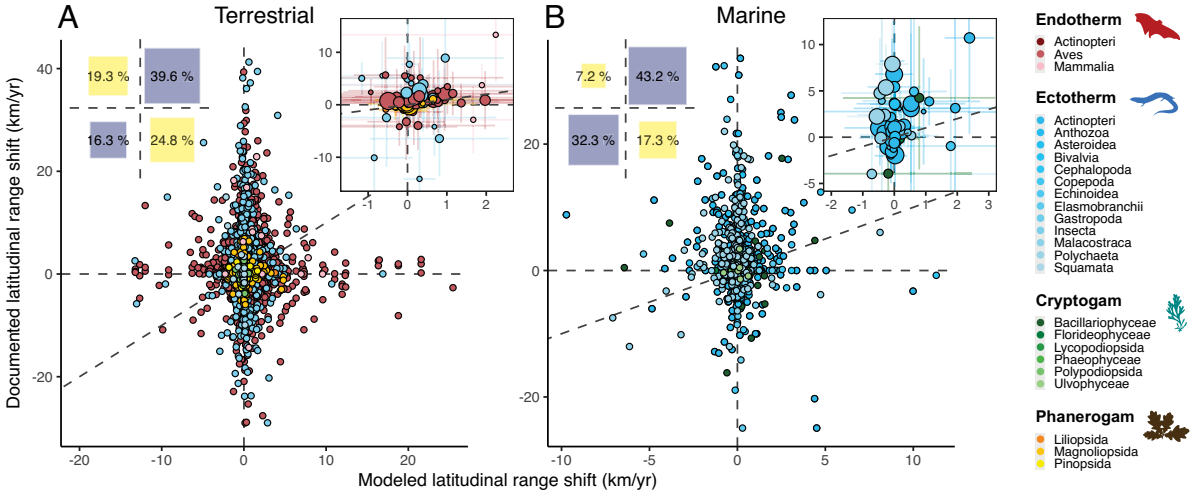

Fig. 2. Relationship between documented and modeled latitudinal range shifts in the terrestrial (*A*) and marine (*B*) realms. The large panels illustrate relationships at the observation-level (i.e., one dot being a single observation of latitudinal range shift for a single species, study area, and range position). The diagonal dashed lines indicate the 1:1 line, where modeled and documented shifts are equal. The *Upper-Left* insets show the percentage of the observation-level data, distributed across four quadrants, where purple boxes represent alignment (i.e., poleward or equatorward) and yellow boxes misalignment in direction (Fig. 1*C*), with box sizes proportional to the number of range shifts per quadrant. The *Upper-Right* insets represent observation-level data aggregated by taxonomic class, study and range position, with dot size proportional to the number of range shift observations used to calculate the mean, and the vertical and horizontal lines crossing each dot indicating the SD on aggregated modeled and documented shifts, respectively. Species are color-coded by taxonomic class into four life-form categories, following Lenoir et al. (36): endotherm and ectotherm animals, and phanerogam and cryptogam plants.

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Although climatic niche models often fail to accurately predict rates at the level of individually documented range shifts, they may better capture aggregated responses across multiple co-occurring species within a given region. We tested this hypothesis by averaging range shifts across studies, taxonomic classes, and range positions, thereby reducing the noise associated with observational-level variability and focusing on general trends. The direction of modeled aggregated latitudinal range shifts aligned with the documented ones in 62.4% of the terrestrial cases and 81.4% of the marine cases, exceeding chance expectations [binomial GLMMs intercepts: terrestrial = 0.51, 95% CI (0.14, 0.87), 1.66 times more likely to align; marine = 1.47, 95% CI (0.82, 2.13), 4.3 times more likely to align]. Moreover, data aggregation yielded significantly positive relationships between modeled and documented shifts in both realms [gaussian GLMMs: terrestrial slope (β) = 0.43, 95% CI (0.35, 0.50); marine β = 2.66, 95% CI (2.35, 2.97); Fig. 2, *Upper Right* Insets). However, in both cases, the slope remained significantly different from 1 ($P < 0.001$; *SI Appendix*, Table S4), indicating that climatic niche models still fail to estimate the magnitude of documented climate-driven range shifts across co-occurring taxa.

Drivers of Mismatch Between Documented and Modeled Latitudinal Range Shift Rates. We tested several ecological and methodological hypotheses to explain the mismatch between documented and modeled latitudinal range shift rates (*SI Appendix*, Table S5). Specifically, we expected that mismatches could arise from high variability in climate suitability (reflecting environmental noise), limited landscape connectivity, or reduced dispersal ability, all of which constrain species' ability to track climate change. For instance, we expected mobility constraints, due to sessile locomotion or low landscape connectivity, to increase lags (i.e., modeled shifts velocities that surpass documented ones), whereas high dispersal ability could generate overshoots (i.e., species shifting ahead of their climatic envelopes). Methodological factors, such as low niche model performance, short study durations, large latitudinal extents, and nonsystematic sampling, were also expected to reduce the accuracy of both modeled and documented shifts and thus amplify mismatches.

To evaluate these hypotheses, we fit GLMMs with a gamma distribution and a log–link function, using the absolute mismatch in latitudinal shift rates (i.e., the absolute value of the difference between documented and modeled range shifts) as the response variable (*Methods*). Predictor variables included: i) niche model performance, ii) study duration, iii) latitudinal extent of the study area, iv) sampling method, v) variability in climate suitability, vi) landscape connectivity, and vii) species locomotion mode (*SI Appendix*, Table S5). To better assess whether these factors differentially influence mismatch, we included a two-way interaction term between each predictor and a categorical variable representing mismatch type (i.e., lag, overshoot, or misalignment; Fig. 1*C*).

Both ecological and methodological factors explained substantial variation in the ability of climatic niche models to predict range shift rates (Figs. 3 and 4 and *SI Appendix*, Table S6) in terrestrial ($R^2_{Marginal} = 0.51$, $R^2_{Conditional} = 0.66$, n = 8,603) and marine ($R^2_{Marginal} = 0.42$, $R^2_{Conditional} = 0.54$, n = 936) realms. In the terrestrial realm, model performance (Boyce index) was negatively associated with lag magnitude [β = −0.23, 95% CI (−0.37, -0.09)] but not with the magnitude of overshoots or misalignments, indicating that more accurate models lead to smaller lags (Fig. 3*A* and *SI Appendix*, Fig. S4*A* and Table S6). Study duration had a negative effect on the mismatch, regardless of its type (i.e., lag, overshoot, or misalignment), with longer studies showing a better match (Fig. 3 *B* and *E* and Fig. S4 *B* and *E* and Table S6). In contrast, variability in climate suitability increased the mismatch, especially so for misalignments [β = −0.18, 95% CI (0.13, 0.23)] and lags [β = 0.32, 95% CI (0.25, 0.38)] (Fig. 3*C* and *SI Appendix*, Fig. S4*C* and Tables S6 and S7). The latitudinal extent of the focal study area also had a positive effect on all mismatch types, with larger mismatches in areas spanning broader latitudinal extents (Fig. 3*D* and *SI Appendix*, Fig. S4*D* and Tables S6 and S7). Landscape connectivity had a modest but negative effect on misalignment [β = −0.65, 95% CI (−0.94, −0.36)], indicating that a higher connectivity improves directional matches (Fig. 3*E* and *SI Appendix*, Fig. S4*E* and Tables S6 and S7). The sampling method used to estimate range shift rates also matters: survey-resurvey

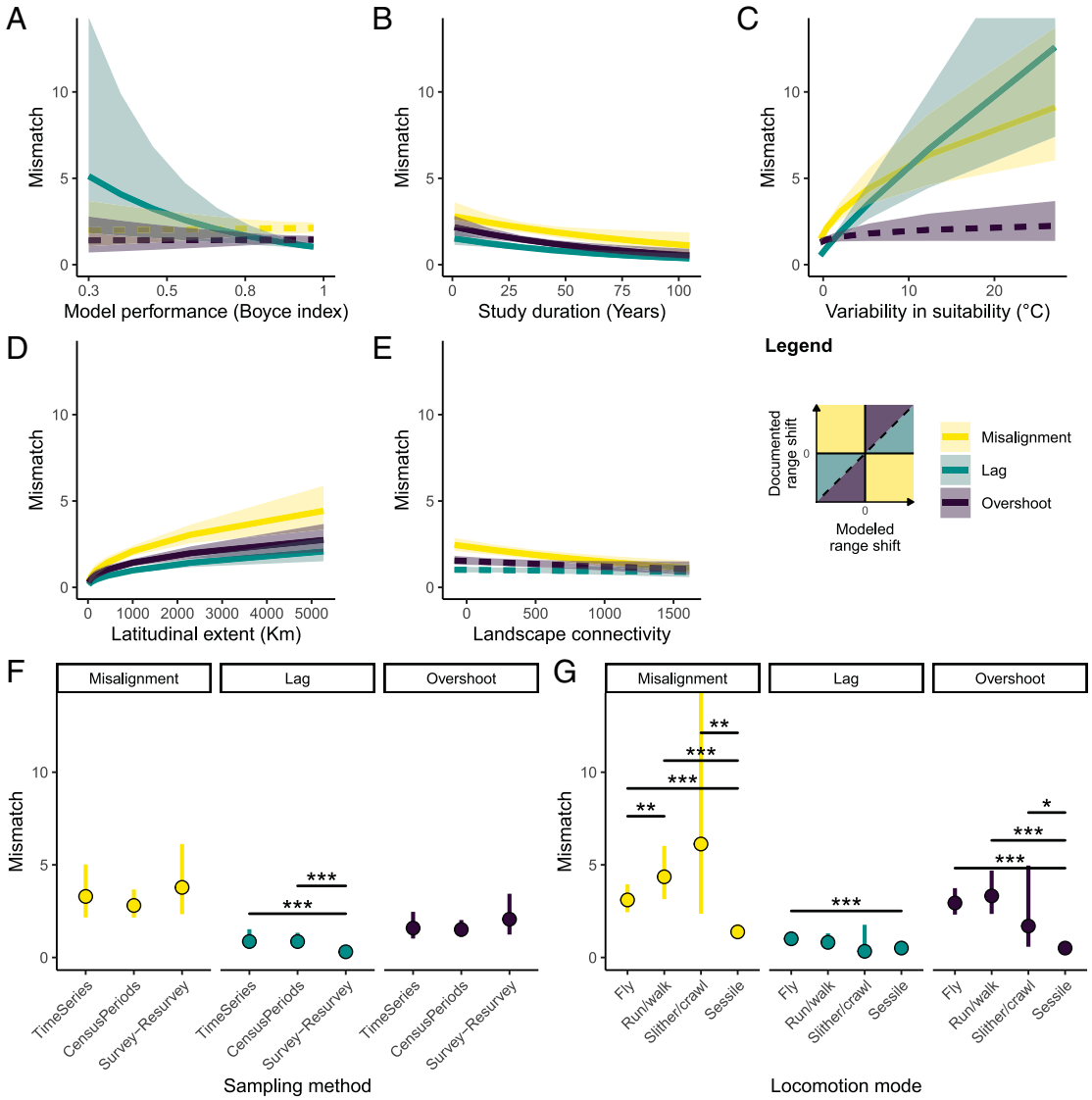

Fig. 3. Drivers of the absolute mismatch between documented and modeled latitudinal range shift rates for terrestrial species according to the nature of the mismatch (lag, overshoot, misalignment) (n = 8,603). Mismatch is the absolute value of the difference between documented and modeled range shifts. Panels show predicted marginal effects of the variables identified as displaying significant associations with the magnitude of the mismatch across three categories: lag (documented shift < modeled shift), overshoot (documented shift > modeled shift), or misalignment (opposite directions between documented and modeled shifts). Associated envelopes with the fitted lines for the continuous (*A*–*E*) predictors are the 95% CI, and solid lines represent significant interaction effects between predictors and the type of mismatch, whereas dashed lines represent nonsignificant interaction effects. Dots and bars for the factor predictors (*F* and *G*) represent marginal means and 95% CI. Lines connecting factor predictors indicate significantly different marginal means: ***$P < 0.001$; **$P < 0.01$; *$P < 0.05$. See *Methods* for further details on the underlying models (GLMMs).

sampling designs reduced lags compared to continuous time series or census periods (Fig. 3*F* and *SI Appendix*, Fig. S4*F* and Table S6), an outcome that was somewhat unexpected. The locomotion mode of terrestrial species influenced their mismatch, with sessile organisms (e.g., plants) showing lower mismatches relative to other locomotion modes (Fig. 3*G* and *SI Appendix*, Fig. S4*G* and Table S6), whereas fly or run/walk species showed greater overshoots and misalignments.

In the marine realm, the effect of study duration, latitudinal extent, sampling method, and locomotion mode varied with mismatch type. Longer study durations reduced overshoots (β = -0.72, 95% CI [-1.02, -0.42]) and misalignments (β = -0.43, 95% CI [-0.83, -0.03]), as observed for terrestrial species, but had no effect on lags (Fig. 4*A* and *SI Appendix*, Fig. S5*A* and Tables S6 and S7). Likewise, larger latitudinal extents increased mismatches regardless of type (Fig. 4*B* and *SI Appendix*, Fig. S5*B* and Tables S6 and S7). In terms of sampling method, continuous time series showed greater lags and overshoots relative to census periods or survey-resurveys (Fig. 4*C* and *SI Appendix*, Fig. S5*C* and Table S6). Last, swimming species displayed greater lags than slithering/crawling ones (Fig. 4*D* and *SI Appendix*, Fig. S5*D* and Table S6). Unlike terrestrial range shifts, we found no evidence that model performance, variability in climate suitability or seascape connectivity affected marine range shift' mismatches (*SI Appendix*, Table S7).

Additionally, we tested whether elevational range shifts help explain the mismatch between documented and modeled latitudinal range shifts, under the expectation that opportunities for elevational adjustments may reduce the need for latitudinal movements. This analysis was based on a subset of terrestrial species (n = 299) with both latitudinal and elevational range shift estimates (n = 512). Including elevational range shifts as a predictor improved model fit (likelihood ratio test: ΔAIC = 6.7, $\chi^2_1 = 8.76$, $P < 0.001$), suggesting that trade-offs between latitudinal and elevational shifts partly

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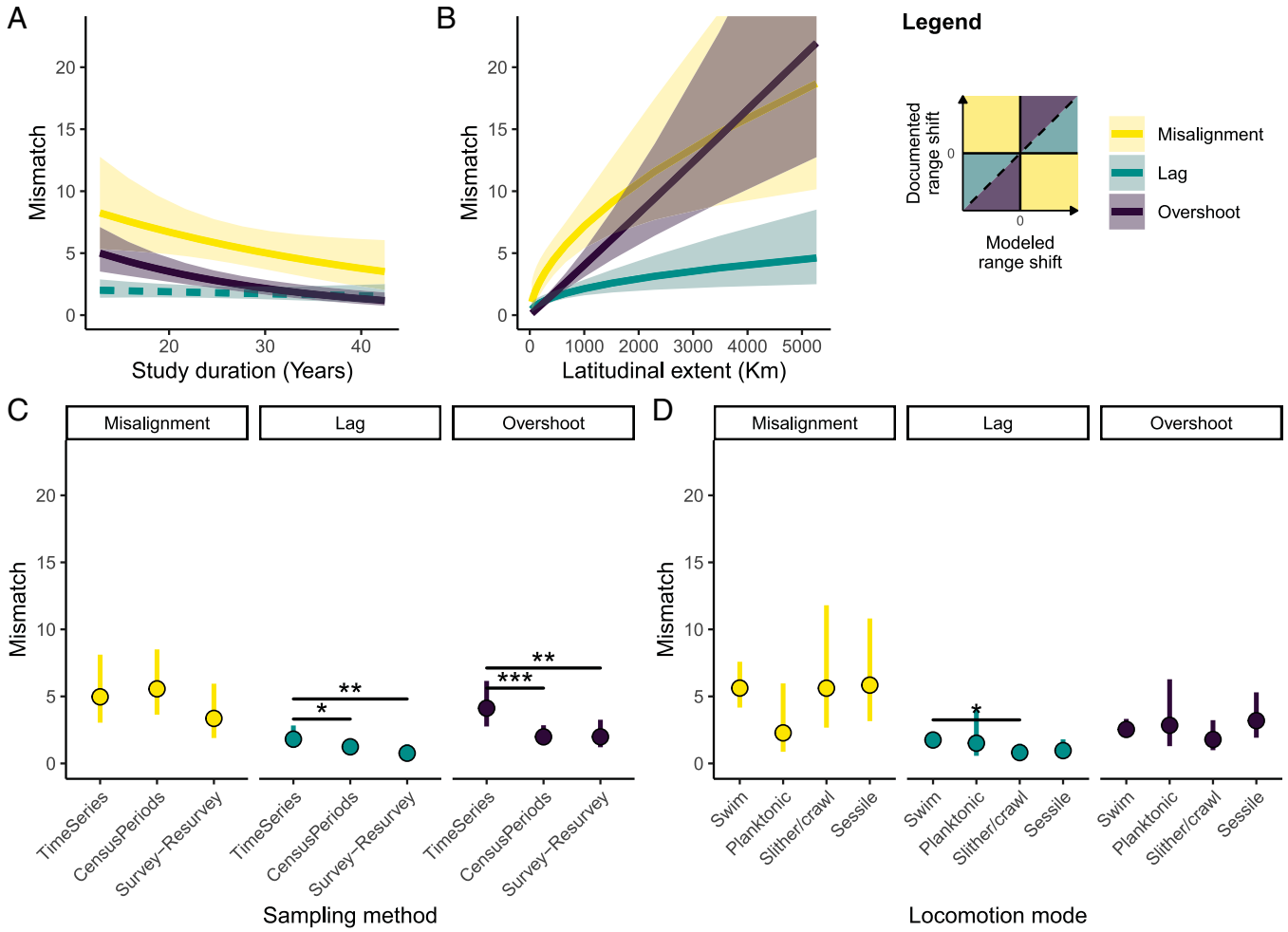

Fig. 4. Drivers of the absolute mismatch between documented and modeled latitudinal range shift rates for marine species according to the nature of the mismatch (lag, overshoot, misalignment) (n = 936). Mismatch is the absolute value of the difference between documented and modeled range shifts. Panels show predicted marginal effects of the variables identified as displaying significant associations with the magnitude of the mismatch across three categories: lag (documented shift < modeled shift), overshoot (documented shift > modeled shift), or misalignment (opposite directions between documented and modeled shifts). Associated envelopes with the fitted lines for the continuous predictors (*A* and *B*) are the 95% CI. Dots and bars for the factor predictors (*C* and *D*) represent marginal means and 95% CI. Lines connecting factor predictors indicate significantly different marginal means: ***$P < 0.01$; **$P < 0.01$; *$P < 0.05$. See *Methods* for further details on the underlying models (GLMMs).

explain the variability in the mismatch between modeled and documented range shift rates (*SI Appendix*, Fig. S6 and Table S8). By contrast, adding absolute latitudinal centroid of each study locality provided no evidence that latitude helps explaining the mismatch in either terrestrial [β = 3e−3, 95% CI (−7e−3, 0.01)] or marine [β = 1e−3, 95% CI (−1.1e−2, 0.01)] realms.

Discussion

Climatic niche models offer a promising framework to forecast biodiversity responses to climate change, yet their predictive reliability remains uncertain across diverse taxonomic and geographic regions. By integrating empirical estimates of species range shifts over the past century (40, 41), we found limited evidence of climatic niche tracking across thousands of terrestrial and marine taxa. This global assessment of the ability of climatic niche models to predict observed range shift rates identifies key factors underlying mismatches between models and observations, offering insights into the mechanisms driving range shift dynamics and guiding the development of more robust predictive tools for biodiversity management under ongoing climate change.

Our results show that although climatic niche models often align with the general direction of documented range shifts, they poorly predict the actual rates of range shifts across landscapes and seascapes. These findings, consistent with previous taxon- or region-specific studies (e.g., 25, 26, 37), reveal broader limitations of forecasts based on climate-only niche models. Contrary to the common expectation that species generally lag behind climate change (42–44), our climatic niche models tend to underestimate range shift rates in both terrestrial and marine realms when aligned in direction, contradicting previous reports that modeled shifts outpace observations (25, 37, 38, 45). For instance, Howard et al. (38) found that European birds shifted their ranges approximately 50% slower than climatic niche predictions. In contrast, and consistent with our results, Huang et al. (24) reported that birds in the contiguous U.S. shifted nearly twice as fast as climatic niche expectations. One possible explanation reconciling whether species lag behind or overshoot model predictions involves the spatial resolution of the underlying climatic data. Huang et al. (24) used climatic data at 5 km resolution, which is much closer to the spatial resolution used for terrestrial species in our study (1 km) than to the spatial resolutions used in other studies reporting more lags (25, 37, 38, 45). Coarse climatic grids smooth local variations over finer-scale refugia—that is, stable microclimatic conditions (30, 46, 47)—potentially inflating estimated species' thermal tolerance when calibrating niche models (48), and yielding overestimated (faster) modeled shifts. Conversely, finer-resolution climatic data should better capture finer-scale refugia (49), thus

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generating smaller predicted range shifts (50). This underscores the importance of accounting for spatial resolution dependencies when interpreting or calibrating SDMs. Because the spatial resolution of the climate data can influence the magnitude of predicted range shifts, it may also alter the inferred mismatch type, shifting from predominantly lags to frequent overshoots. Moreover, as different taxonomic groups perceive and respond to environmental heterogeneity at different spatial resolutions (51), the optimal resolution of the underlying climatic data used in climatic niche models should be taxon-specific (52). This, however, complicates attempts to generalize model performance across biodiversity datasets covering a diverse array of taxonomic groups.

The finding that observed range shift rates often outpace modeled ones, when aligned in direction, is particularly striking given that our climatic niche models assume no dispersal limitation (i.e., species are presumed able to access all climatically suitable areas). This challenges another key assumption of climatic niche models: the equilibrium assumption (28). When species are in disequilibrium with their optimal climatic conditions (e.g., due to biotic interactions, physical or anthropogenic barriers) they may fail to fully occupy their fundamental niche. Such niche truncation, where the realized niche is restricted by the available environmental space (33, 53), can yield overly conservative projections of range expansion (54), leading to underestimated range shift rates and overestimated climate vulnerability (55). Although we fitted our models across the full potential accessible area of each species to minimize geographic truncation (56), and used a sample-with-data approach to limit the influence of emerging nonanalogue climates over time (57), residual truncation may still arise from spatial bias in occurrence records (33). Another limitation is that projections were made over continuous climatic surfaces without constraining predictions to suitable habitat types (e.g., land cover masks), which may inflate the extent of climatically suitable areas while diluting projected shift velocities, particularly for species with patchy distributions or narrow habitat requirements. Together, these factors underscore the limitations of correlative climate-only models in forecasting species redistribution.

Our results corroborate earlier evidence of stronger climate tracking in the oceans than on land (36, 58), likely reflecting fundamental differences in physical environments and organismal responses (59). The direction of documented latitudinal range shifts (i.e., poleward *vs.* equatorward) aligned more closely with climatic niche predictions in marine than in terrestrial systems. This pattern may reflect more uniform environmental gradients and the predominance of ectothermy in marine taxa, which increases sensitivity to temperature variation (59, 60). In contrast, terrestrial species face greater constraints from geographical barriers (e.g., topography, land use, and habitat fragmentation) that impede redistribution in context-dependent ways. For instance, we found that higher landscape connectivity reduced mismatches, particularly misalignments, in terrestrial systems, whereas seascape connectivity showed no effect in marine systems. Connectivity likely facilitates movement on land, enabling species to track climate shifts more accurately (61, 62). Nevertheless, differences in the spatial resolution of terrestrial (1 km) and marine (25 km) datasets may also contribute to the degree of mismatch, as spatial scale affects both model accuracy and range-shift detectability. Consequently, our realm-specific results could be interpreted as scale-dependent patterns rather than direct quantitative contrasts, and future work using harmonized spatial resolutions is needed to disentangle biological from methodological drivers of these differences.

Species' dispersal ability modulates how well organisms track shifting climates, though its effect on mismatch is more nuanced than previously assumed (14, 63, 64). As expected, mobile terrestrial species (e.g., flying, running/walking dispersal modes) more often overshot their modeled shifts than sessile terrestrial species (e.g., trees). However, in addition to the very strong tendency to overshoot more than sessile organisms, flying organisms simultaneously exhibited a slight tendency to lag more often than their sessile counterparts, a counterintuitive result that challenges assumptions of faster responses among more mobile taxa. This less conspicuous pattern to lag, as opposed to overshoot, may reflect the greater behavioral flexibility among flying organisms, which can temporarily occupy novel habitats or seek shelter in cooler microclimates within their home range, thereby delaying directional shifts relative to model expectations. Similarly, in marine systems, swimming species showed more lags than less mobile crawling species. This may occur because latitudinal shifts, as measured here, overlook vertical movements within the water column that some marine taxa can use to remain within their thermal niches. For example, many fish species track favorable temperatures by adjusting depth rather than latitude (65). In contrast, crawling or slithering species are closely tied to the seafloor, responding more gradually and predictably to horizontal changes in sea surface temperature—the axis captured in our analysis—owing to their limited capacity for vertical movement along the water column. Furthermore, the closer alignment between documented range shifts and model predictions for these less mobile taxa may also reflect larval dispersal rather than adult movement, as population gains in suitable areas and losses elsewhere can create the appearance of spatial tracking (66). Additional unmeasured traits, including habitat specialization, ecological breadth, life-history, and biotic interactions, may also underlie these mismatches and merit further investigation. Elucidating how climate exposure, dispersal processes, and species-specific traits jointly shape range dynamics remains a key frontier for future research (14).

Interannual variability in climatic suitability further mediates the alignment between modeled and documented range shifts in terrestrial systems, contributing to both lags and mismatches. Marine species appear less affected, potentially due to the buffering capacity of oceanic thermal environments and enhanced dispersal via ocean currents (67). On land, strong interannual fluctuations can induce transient population responses, especially in temperate regions, leading species to adjust behaviorally (e.g., seeking shelters in cooler microhabitats or shifting daily activity period to avoid thermal stress) or phenologically (e.g., changes in migration or reproductive timing) rather than geographically (9, 68), thereby contributing to the observed lags and misalignments relative to model expectations. Interannual variability may also explain why mismatches were generally lower in studies spanning longer timeframes: extending the temporal coverage smooths short-term population responses and reveals clearer directional trends, thereby improving alignment between modeled and observed range shifts (35, 69). Accordingly, higher uncertainties in estimated range shift rates are expected in short-duration studies (8), underscoring both the value of long-term monitoring programs, typically spanning 50 to 70 y, and the need for caution when interpreting results from short-term datasets that may be disproportionately influenced by stochastic events or demographic inertia.

The latitudinal extent of the study area substantially influenced the accuracy of range shift predictions (35, 36). Small study areas with limited latitudinal extents should impose geometric constraints on both modeled and observed range shifts (70), thus decreasing the chances for mismatches to arise. Consistent with this expectation, we found that mismatches increased with latitudinal extent, particularly in the form of greater misalignments and

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overshoots. Greater environmental heterogeneity and climatic variability across broader spatial extents likely result in complex species' responses that climatic niche models may not fully capture. Climatic niche models estimate range shifts by projecting changes in climatic envelopes across continuous spatial surfaces, effectively smoothing over environmental gradients and reducing sensitivity to site-specific outliers. In contrast, observed range shifts are typically derived from localized (i.e., site-specific) observations that are more sensitive to sampling artifacts, particularly at the leading and trailing edges of species' ranges. Smaller study areas may reduce the impact of these uncertainties, as localized observations are more likely to reflect the spatial resolution and extent at which the models operate.

The sampling characteristics of the underlying species observations can also influence the uncertainty in range shift estimates (35). We modeled range shifts by regressing latitudinal positions of range limits and centroids against time, a method sensitive to climate variability (i.e., high interannual variability increases scatter around the trend, inflating residual variance and reducing slope precision). This approach closely mirrors the continuous nature of time series data, capturing year-to-year variation rather than the discrete temporal intervals typical of survey–resurvey or census studies. Therefore, we expected modeled shifts to more closely align documented shifts from continuous time series studies. Surprisingly, documented range shifts from time series studies were more likely to lag behind modeled shifts than those from survey-resurvey studies. While time series provide finer temporal resolution, through continuous surveys spanning multiple years, they may obscure long-term patterns by emphasizing short-term fluctuations or lagged responses, reducing the signal-to-noise ratio in empirically documented shifts (69). In contrast, survey-resurvey studies, with samplings often separated by decades, may allow cumulative climatic effects and associated biological responses to emerge, yielding observed shifts more consistent with climatic niche expectations.

Although our analyses focused on latitudinal range shifts, elevational gradients represent another important dimension for species redistribution that may explain part of the apparent mismatches we found between documented and modeled latitudinal range shifts. With a typical lapse rate of 5-6 °C per km, a 1 km upslope shift corresponds roughly to a 1,000 km poleward shift. Based on a subset of species for which both range shift dimensions were available, we found evidence that some of the mismatches between observations and predictions may indeed reflect trade-offs between latitudinal and elevational range shifts. This finding underscores the importance of accounting for topography when assessing species' responses to climate change (71), including in climatic niche models (50). Similar trade-offs may also exist between latitudinal and depth shifts for marine species (65) as well as between elevational and vertical shifts for tropical forest species (72), and the three-dimensional structure of habitats should also be considered to better capture redistribution processes below the sea surface or below the canopy.

Our dataset on latitudinal range shifts is biased toward temperate regions (*SI Appendix*, Fig. S1), where temperature gradients along the latitudinal dimension are steeper than in the tropics. In a recent synthesis, Colwell & Feeley (73) showed that latitudinal range shifts are rare, or at least poorly documented, in the tropics, despite a comparable representation of elevational range shifts across regions. In tropical lowlands, precipitation gradients can play a dominant role in shaping species' distributions (11, 74), and changes in rainfall patterns can even drive equatorward movements (15). Yet, we found no evidence that absolute latitude helps explain the mismatch between documented and modeled latitudinal shifts in either realm. Nevertheless, expanding tropical coverage, where species' thermal sensitivities differ markedly from temperate regions (75), could help disentangle the roles of climatic exposure, physiological responses, and ecological traits in shaping species redistribution under climate change (14).

Although climatic niche models provide useful predictions, transferability remains a major concern, as high accuracy metrics (e.g., Boyce, TSS) do not always guarantee reliable forecasts under changing climates (34). Climatic niche models were not originally designed for forecasting range shifts, and they lack the process-based mechanisms required to simulate dispersal, adaptation, or biotic interactions. Improving forecasts will require integrating correlative models with mechanistic insights, such as experimentally derived physiological thresholds, dynamic dispersal kernels, landscape resistance, and multitrophic interactions (76–78). This is particularly urgent for terrestrial systems, where mismatches are more pronounced. However, increasing model realism must be paired with empirical validation through long-term, spatially explicit monitoring that aligns with climatic niche outputs. Continued global efforts to compare climate-only forecasts with observed range shifts, as in this study, are essential to track progress and ensure that model refinements translate into meaningful gains in understanding and forecasting biodiversity redistribution under climate change.

Methods

Documented Range Shift Data. We compiled empirical observations of species latitudinal range shifts from two key geodatabases: Bioshifts (40) and CoRE (41). Both provide compilations of range shift estimates (km/y) from the scientific literature, with metadata on methodological aspects and digitized study areas polygons of the original studies. We merged these databases by harmonizing taxonomy, standardizing units, and metadata, and removing duplicate entries to create a unified dataset of latitudinal range shifts across marine and terrestrial taxa. From the original set of range shift estimates, we excluded species identified only at the genus level, nonsignificant range shifts likely to introduce noise, and range shifts documented beyond the range of the available climatic datasets (i.e., pre-1902 for terrestrial and pre-1959 for marine species; see *Climate data*). We also removed freshwater fishes and marine birds, as their responses may be better captured by alternative environmental variables (e.g., stream or lake temperatures for freshwater fish and air temperature above the ocean for marine birds). To allow comparison between the marine and terrestrial realm, we focused on range shifts along the latitudinal gradient, and included shifts observed at the centroid, leading edge, and trailing edge. This first filtering procedure resulted in a selection of 4,341 candidate species (3,633 terrestrial and 708 marine).

Modeled Range Shifts Data.

Climate data. For terrestrial species, we obtained monthly maximum and minimum temperatures, as well as the total monthly rainfall, from January 1901 to December 2019 from CHELSAcruts, an extension of the CHELSA database (79, 80). Data were sourced at a spatial resolution of 0.008° × 0.008° (~1 km × 1 km), consisting of 1,428 monthly layers. For marine species, monthly sea surface temperature (SST) from January 1958 to December 2020 were obtained from the Ocean Reanalysis System 5 (ORAS5; https://data.marine.copernicus.eu) at a spatial resolution of 0.25° × 0.25° (~25 km × 25 km). These SST data consist of gap-free monthly means, free of diurnal variability (81).

Species occurrence data. Occurrence records were retrieved from the Global Biodiversity Information Facility (GBIF) database (https://www.gbif.org/, see Dataset S1 for a list of DOIs from the downloaded sets of occurrences) in November 2022 via R v4.2.2 using the function *occ_search* from the R package *rgbif* (82). We filtered all occurrence records from a given species by keeping only occurrences within the temporal window of the environmental data (see *Climate data*), describing a presence ("occurrenceStatus = PRESENT"), with latitude/longitude reference ("hasCoordinate = TRUE"), without spatial issues ("hasGeospatialIssue = FALSE"), and for which the field "basisOfRecord" was "HUMAN_OBSERVATION", "OBSERVATION", or "OCCURRENCE". We avoided further potential spatial issues

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using the function *clean_coordinates* from the R package *CoordinateCleaner* (83), excluding coordinates near capitals (10 km radius), country centroids (1 km radius), GBIF headquarters (100 km radius), biodiversity institutions (100 m radius), or with identical latitude and longitude values. We additionally removed terrestrial records on ocean and marine records on land.

Occurrence records were defined as unique combinations of latitude, longitude, and date (i.e., month and year) for each species. Duplicates were removed at the spatial resolution of the climate data (*Climate data*). Given potential imperfect coverage of species distributions by GBIF, we did not attempt to remove geographical outliers (i.e., records located far from the main cluster of observations) as they may contain valuable information on species' climatic niches. We only fit climatic niche models for species with more than 30 occurrence records. In order to minimize singularity issues in our statistical models (*Assessing the Alignment Between Documented and Modeled Range Shifts*), we required a minimum of 5 range shift observations per combination of taxonomic class and range parameter. This threshold ensures adequate variation in our data, allowing more accurate parameter estimation and thus more robust statistical inferences. This filtering yielded 3,951 candidate species (3,380 terrestrial and 571 marine).

Pseudo-absences. Because true absence data are rarely available, we generated pseudo-absences within hypothesized accessible areas (56, 84). We defined accessible (or background) areas as all grid cells in ecoregions (i.e., areas sharing distinct sets of species; 79) intersecting species' occurrence records. We used ecoregions delineated by the World Wildlife Fund (WWF) (85) and the ecoregions and provinces delineated by the Marine Ecoregions of the World (86) to define background areas for terrestrial and marine species, respectively.

For each species, we randomly sampled the same amount of pseudo-absences as occurrence records, ensuring no pseudo-absence overlapped with an occupied grid cell (54, 81). We assigned random pseudo-dates for each pseudo-absence, constrained by the temporal range of the species' occurrences. We adopted a one-to-one presence and pseudo-absence ratio because it performs well (or better) than using a large number of pseudo-absences when fitting our niche-based models, especially when equal weights are applied, according to the comparative analysis of Barbet-Massin et al. (87). See *Sensitivity analyses* for additional tests of the influence of our modeling decisions on model results.

Climatic niche models. We modeled species distributions using a sample-with-data approach, matching each presence or pseudo-absence record with environmental conditions at its specific location and date. This allowed us to generate fine-tuned niche models reflecting the climatic conditions experienced by species during the temporal extent of the original empirical range shift estimates. For each species, we aggregated monthly temperature variables (mean, minimum, maximum, and SD) over the 12-month period preceding each individual occurrence or pseudo-absence record. Terrestrial species also included four precipitation metrics (cumulative, minimum, maximum, and coefficient of variation). Training models with data from the 12-month preceding each record rather than relying on long-term climate averages allowed us to account for the effect of variability in suitability in climatic conditions on species distributions. Prior work has shown that using climatic conditions preceding the year of occurrence record consistently improves predictive performance of niche-based models compared to long-term climate averages (39, 88).

Collinearity among predictor variables can cause instability in parameter estimation in correlative models (89). We avoided this issue using species-specific uncorrelated and orthogonal principal components (PCs) (90). For each species, we first randomly selected 100,000 geographical coordinates at its background area (*Pseudo-absences*), then assigned random dates to these random locations constrained by the temporal extent of the occurrence records. Next, we extracted environmental data from these random locations tagged in time, and used this dataset to compute the species-specific environmental space, using a principal components analysis (PCA). Then, we extracted the minimum set of PCs that accumulated more than 95% of the total variability in the full dataset. We projected into this PCA space the set of environmental conditions associated with the occurrence records and pseudo-absences used to fit the niche models of the focal species, as well as the interannual time series of environmental conditions across each study area, used to project the niche models.

We fitted climatic niche models using an ensemble modeling approach based on four algorithms using the R package *biomod2* (91) v4.2.6.1 default parameters (Dataset S2): generalized linear models (GLM), generalized additive models (GAM), boosted regression trees (BRT), and maximum entropy (MaxEnt). Models were tuned using a prevalence of 0.5, so that presences and absences were weighted equally. Model performance was assessed using true skill statistic (TSS) and the Boyce index (92) via a threefold cross-validation procedure, keeping 2 of the threefolds (70% of data) for model training and the remaining fold (30% of the data) for testing and repeating this three times so that each fold is used at least once for model testing. TSS evaluates the alignment between observed binary outcomes (e.g., presences and absences) and model predictions, whereas the Boyce index focuses only on presence data, comparing the model predicted presences against expected presences in classes of prediction probability. Both metrics range from −1 to +1, with higher values indicating better model performance. The 12 resulting models (4 algorithms × 3 different folds per algorithm) per species were assembled using a weighted-mean approach based on TSS (93). Finally, each ensemble model was projected across the study area where each documented range shift of the focal species had been estimated, and for each consecutive year within the time period covered by the original study. By doing so, we produced yearly time series of suitable climatic space that were subsequently used to estimate modeled range shifts (*Estimating modeled range shifts*) as well as the variability in climate suitability (*Assessing the Mismatch Between Documented and Modeled Range Shifts*) for each species × study area × range parameter combination.

The 1 km × 1 km spatial resolution of the terrestrial models, combined with yearly temporal projections, resulted in approximately 85 terabytes of data and required a wall time of approximately 200 d (computation time distributed across many cores in a high-performance computing facility), justifying the need for substantial computational resources to support fine-scale biodiversity modeling.

Estimating modeled range shifts. Based on the yearly suitability maps derived from the climatic niche models, we calculated the modeled latitudinal range shifts for three different range parameters (centroid, leading edge, and trailing edge) according to the information associated with documented range shifts. Following the BioShifts and CoRE databases, we assumed that higher latitudes in both the northern and southern hemisphere are colder and represent the leading edge of the species range, while lower latitudes are warmer and represent the trailing edge of the range. To enable direct comparison with the documented range shifts, we focused on the latitudinal component of modeled range shift. To estimate the modeled range shifts at the centroid, we first computed the yearly centroid positions as the mean grid cell latitudinal coordinates within the study area weighted by the climatic suitability for each year of the time series. To estimate the modeled range shifts at the leading and trailing edges, we used a similar approach but relying on weighted percentiles, instead of weighted means, using the 75th and 25th percentile of the grid cell latitudinal coordinates in the northern hemisphere and the 25th and 75th percentile of the grid cell latitudinal coordinates in the southern hemisphere. These time series of range positions were then regressed against their corresponding years using simple linear regressions, with slopes representing modeled latitudinal shifts (km/y). All climatic niche predictions were converted to an Eckert cylindrical equal-area projection before any range shift calculation to ensure uniform grid-cell distances.

To assess whether modeled shifts were primarily oriented along the latitudinal axis, we also computed longitudinal shifts using the same approach as for latitudinal shifts and combined both components to obtain total shift magnitudes calculated as the Euclidean combination of the latitudinal and longitudinal components. We quantified the contribution of the latitudinal component as the proportion of the total shift attributable to latitude (values close to 1 indicate predominantly latitudinal shifts). In addition, we calculated the angle of each shift in degrees based on the latitudinal and longitudinal components. This approach allowed us to characterize the directional orientation of each modeled shift. Angles were computed such that 0° corresponds to a northward shift, increasing clockwise, with 90° indicating eastward and 180° indicating southward movement.

Assessing the Alignment Between Documented and Modeled Range Shifts. We fitted generalized linear mixed-effects models (GLMMs) to evaluate whether modeled range shifts from our climatic niche models accurately predict the documented range shifts as reported in the scientific literature. To account for nonindependence of observations (e.g., range shifts reported for different species within the same study being influenced by shared abiotic or methodological factors, range shifts for species within the same taxonomic class being more similar due to shared species-specific sensitivities, or range shifts for a given

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range parameter being driven by the same biological processes), we included a nested random intercept term with the study ID (a factor variable with 72 and 44 levels for the terrestrial and marine realm, respectively) within the taxonomic class (a factor variable with 9 and 15 levels for the terrestrial and marine system, respectively) within range position (a factor variable with three levels: trailing edge, centroid, and leading edge).

First, to test whether the direction of modeled and documented range shifts aligned, we fitted an intercept-only binomial GLMM (no fixed effect) on a binary response variable of alignment (coded 1) *vs.* misalignment (coded 0) in range shift direction. The intercept, expressed as a log-odd ratio, was exponentiated to convert it to odds. Next, to test whether rates of modeled and documented range shifts were positively related, indicating coupling between predictions and observations, we fitted a GLMM with documented range shifts as the response variable and modeled shift as the sole fixed-effect variable, specifying a Gaussian error structure. To evaluate whether the magnitude of modeled rates matched or deviated from documented magnitudes, we set the null hypothesis of the slope estimate to 1 instead of 0. To test for consistency between modeled and documented range shifts across a set of co-occurring species, we aggregated range shift rates by Study ID, taxonomic class, and range parameter. Specifically, we modeled the mean of aggregated documented shift rates as a response, with the mean of aggregated modeled rates as the sole fixed effect, and used the number of species within a given study combination as weights. For these GLMMs, nested random intercepts included taxonomic class within range position.

All analyses were performed separately for terrestrial and marine species due to differences in environmental datasets and resolutions. We evaluated the goodness of fit of the GLMMs using the variance explained by each model's fixed effects (marginal R^2) and combination of fixed and random effects (conditional R^2), following Nakagawa et al. (94). All GLMMs were fitted with the R package *glmmTMB* (95).

Identifying the Drivers of the Mismatch Between Documented and Modeled Range Shifts. We investigated factors influencing the absolute mismatch between documented and modeled range shifts using five categories of predictors variables (*SI Appendix*, Table S5): i) model performance, ii) spatiotemporal resolution of the documented range shifts, iii) variability in climate suitability, iv) landscape connectivity and v) species traits related to ability to move and life form categories.

(i) As a metric of model performance, we calculated the Boyce index (mean across the threefold cross validation). We expected that poorly performing models, resulting from niche truncation and/or missing key ecological variables, under or overestimate actual species distributions (17), generate inaccurate range shift predictions and thus higher mismatch.

(ii) To capture the spatiotemporal resolution of the documented range shifts, we extracted four different metrics from the BioShifts and CoRE databases: number of sampling periods, study duration, sampling method (time series: annual or decadal species records, surveyed/resurveyed: two annual species records separated in time, or census periods: two periods of species records composed of multiple years and separated in time), and latitudinal extent of the study area (from the northernmost to the southernmost points). We hypothesized that range shifts estimated from limited temporal observations (i.e., surveyed/resurveyed) are more sensitive to atypical years as compared to extensive time series or observations over multiple sampling periods, which are expected to better reflect long-term species' range dynamics (69). Similarly, range shifts estimated from studies spanning large latitudinal extents are less likely to align with niche-based expectations because broader gradients relax geometric constraints and increase habitat heterogeneity, both natural and anthropogenic, thereby reducing predictability.

(iii) To capture the degree of temporal variability in species' suitable climatic niche space, we used the SE of the regression slope tracking the latitudinal position of the focal range parameter (centroid, leading edge, or trailing edge) against time (years). We expected that temporal variability in (climatic) suitability, reflecting short-term climatic fluctuations that pushes and pulls species distributions, would reduce the predictability of observed range shifts. Accordingly, species exhibiting low temporal variability in their suitable climatic space should display more consistent modeled trends and therefore a closer match between documented and modeled range shifts (96).

(iv) To assess the influence of landscape or seascape connectivity on organism movement, we applied a circuit-theoretic approach (97), based on two inputs: landscape resistance, the degree to which landscape features restrict movement, and source weights, the likelihood that movement originates from a given location. Using these inputs, we calculated the current flow metric (97), which quantifies permeability to movement. For terrestrial systems, we modeled connectivity globally at 1 km resolution using Omniscape, where landscape resistance and source strength were informed by landscape naturalness (98), a proxy for human modification. Movements were allowed only between locations with similar land cover (e.g., forest, wetland, grassland) and decadal temperature within 1.11 °C, assuming a maximum dispersal distance of 200 km. For marine systems, we used the analogous current flow metric from Petsas et al. (99), representing the probability that thermal-analog trajectories intersect a pixel. To obtain species- and study-specific connectivity values, we averaged the connectivity index across all grid cells of each study area weighted by species climatic suitability at baseline (i.e., year 1). For range edges, we extracted mean connectivity within the 75th–95th (leading) and 5th–25th (trailing) suitability-weighted latitudinal percentiles in the northern hemisphere, reversing percentiles in the Southern Hemisphere. We expected that poorly connected landscapes impede expansion and accelerate contraction (100), increasing the mismatch between modeled and documented range shifts.

(v) Species' locomotion mode was classified into: flying, running/walking, slithering/crawling, sessile, swimming, or planktonic. We expected species with limited mobility to show smaller range expansions and thus greater mismatch between documented and modeled range shifts (14). To account for broad physiological differences in temperature sensitivity, we further categorized taxa into four life form groups following Lenoir et al. (36): endotherms, ectotherms, phanerogams, cryptogams.

To test these hypotheses, we fitted separate GLMMs for terrestrial and marine systems using the absolute mismatch between modeled and documented range shift rates as the response variable, with a Gamma error structure and a log link function. We focused on absolute mismatch values because we were interested in the difference between shift rates, but also included an interaction term representing mismatch type (i.e., overshoot = documented > modeled range shifts; lag = documented < modeled range shifts; misalignment = documented and modeled range shifts differ in direction). This framework allowed us to capture the effects of the hypothesized drivers on both the direction and magnitude of the range shifts. All predictor variables were included as additive fixed effects. To address multicollinearity (VIF > 4), we excluded two variables, i.e., number of sampling periods and life form, from both terrestrial and marine models. To linearize relationships, we log-transformed latitudinal extent and connectivity, and log+1 transformed *variability in suitability*. All continuous predictors were standardized to allow comparison of effect sizes and backtransformed for plotting. We used a nested random intercept structure (1 | PositionID/ClassID/StudyID) consistent with previous models. Using the *emtrends* function from the R package *emmeans* (101), we estimated the slopes of continuous predictors for each mismatch type and assessed pairwise contrasts of estimated marginal means to test differences among categorical levels. Model fit was evaluated using marginal and conditional R^2 following Nakagawa et al. (94). All GLMMs were fitted using the R package *glmmTMB* (95).

Sensitivity Analyses. We conducted a series of supplementary analyses to evaluate the robustness of our results to other ecological and modeling factors that may be relevant to the comparison between modeled and documented range shifts.

First, we tested whether the relationship between modeled and documented range-shift rates depends on the alignment of their directions by incorporating an interaction term in the GLMMs with three alignment categories: poleward alignment, equatorward alignment, and misalignment. Climatic niche models predicted documented rates more reliably when shift directions were aligned, particularly for equatorward shifts (*SI Appendix*, Fig. S7 and Table S8). To evaluate whether this improvement could arise by chance, we implemented a null-model approach in which alignment labels were randomly permuted across observations 1,000 times and compared observed conditional and marginal R^2 values with those from permuted models. Observed R^2 values were consistently higher than null expectations (P-values < 0.001), indicating that the improvement was not a statistical artifact but reflects a meaningful relationship between alignment direction and predictive accuracy.

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We also assessed sensitivity to how range positions were defined by repeating analyses using the 5th and 95th percentiles (representing trailing and leading edges in the northern hemisphere, and the reverse in the Southern Hemisphere). These produced qualitatively similar results (*SI Appendix*, Fig. S8 and Tables S9 and S10), confirming that our conclusions are robust to this methodological choice. Accordingly, we present in the main text results based on the 25th and 75th percentiles.

Although our main analyses focused on latitudinal range shifts, elevational gradients represent another key dimension of species redistribution. We therefore evaluated the role of elevation on the mismatch between modeled and documented range-shift rates using a subset of terrestrial species (n = 299) for which both latitudinal and elevational range-shift estimates were available (n = 512). We compared models with and without elevation range shifts as a predictor of the mismatch using likelihood ratio tests based on changes in AIC and deviance (*SI Appendix*, Fig. S6). Likewise, we assessed, separately for terrestrial and marine datasets, whether including study latitudinal position (based on the study-area centroid) improved model fit.

We further evaluated robustness to model performance by rerunning analyses on a subset of species with well-performing climatic niche models (TSS > 0.5; species = 2,738; range shifts = 7,330). As this yielded qualitatively similar results in both realms, we report full-dataset results in the main text and those from the filtered subset in *SI Appendix*, Figs. S9 and S10.

Finally, while previous studies have adopted a range of defensible modeling choices, the sensitivity of results to these choices is often unexplored (102). We therefore evaluated the robustness of modeled range shifts using an alternative SDM workflow that varied three key decisions: the spatial thinning of occurrences, the number of pseudo-absences, and the temporal allocation of pseudo-dates (see Supporting Text). Occurrences were spatially thinned at 5 km using the R package *GeoThinneR* (103), applied only to terrestrial species because the coarser spatial resolution of marine environmental data yields inherently sparse and weakly clustered occurrences. We selected 1,000 pseudo-absences for species with fewer than 1,000 occurrences and 10,000 for species with more than 1,000 occurrences. Instead of assigning pseudo-dates at random, pseudo-dates were sampled in proportion to the annual distribution of occurrences, preserving the temporal structure of the presence data. All other modeling decisions were unchanged. Analyses were conducted on a random 10% subset of terrestrial (N = 303) and marine (N = 54) species with fewer than 10,000 occurrences, balancing computational feasibility with representativeness across ecological and biogeographic contexts. We then compared the original and alternative workflows in terms of model performance (TSS and Boyce index) and the degree of mismatch between documented and modeled range shifts (*SI Appendix*, Figs. S11-S13).

Data, Materials, and Software Availability. All study data are included in the article and/or *SI Appendix*. All code used to reproduce the analyses reported in this study is publicly available at https://github.com/Bioshifts/Exposure-SDM (104).

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Author affiliations: [a]Centre de Synthèse et d'Analyse sur la Biodiversité, Fondation pour la Recherche sur la Biodiversité, Montpellier 34000, France; [b]Departamento de Ecologia, Universidade Federal do Rio Grande do Norte, Natal 59072-970, Brazil; [c]Centre de Recherche sur la Biodiversité et l'Environnement (UMR5300), CNRS, Institut de Recherche pour le Développement, Institut National Polytechnique, Université Toulouse 3 - Paul Sabatier, Toulouse 31077, France; [d]Department of Ecology and Evolutionary Biology, University of California, Santa Cruz, CA 95060; [e]Centre for Marine Socioecology, University of Tasmania, Hobart 7000, Australia; [f]Institute for Marine and Antarctic Studies, University of Tasmania, Hobart 7000, Australia; [g]Department of Wildlife Ecology and Conservation, University of Florida, Gainesville, FL 32611; [h]Red de Biología Evolutiva, Instituto de Ecología A.C., Xelapa 91073, Mexico; [i]Department of Life Sciences, National Cheng Kung University, Tainan 701, Taiwan; [j]Department of Biology, McGill University, Montreal, QC H3A 1B1, Canada; [k]National Climate Adaptation Science Center, U.S. Geological Survey, Reston, VA 20192; [l]School of Natural Resources, University of Tennessee, Knoxville, TN 37996; [m]School of Biological Sciences, University of Aberdeen, Aberdeen AB24 2TZ, United Kingdom; [n]Graduate Program in Ecology and Evolution, School of Environmental and Biological Sciences, Rutgers, The State University of New Jersey, New Brunswick, NJ 08901-8551; [o]Department of Biology, Case Western Reserve University, Cleveland, OH 44106; [p]UMR CNRS 7058, Ecologie et Dynamique des Systèmes Anthropisés, Université de Picardie Jules Verne, Amiens 80000, France; and [q]Conservation Science Partners, Inc., Truckee, CA 96161

Author contributions: B.F.O., R.B., M.L.P., N.C., B.W.W., B.R.S., F.V., G.G., G.T.P., I.-C.C., J.A.L., J.S., J.M., J.R., L.M.T., L.T.L., M.A.R., N.A.M., R.M.W.J.B., S.E.D., S.R.W., J.L., and L.C. designed research; B.F.O., R.B., M.L.P., J.L., and L.C. performed research; B.F.O., R.B., M.L.P., N.C., J.A.B., J.L., and L.C. analyzed data; and B.F.O., R.B., M.L.P., B.W.W., B.R.S., F.V., G.G., G.T.P., I.-C.C., J.A.B., J.A.L., J.S., J.M., J.R., L.M.T., L.T.L., M.A.R., N.A.M., R.M.W.J.B., S.E.D., S.R.W., J.L., and L.C. wrote the paper.

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